

Biology — Exam 13 — Chapters 1-7 (Full Syllabus), Detailed Marking Report

Seemab Ali | Class 9 | Annual Examination 2026 | Federal Board (FBISE) SSC-I | Paper Version 1 | Attempt date: 2026-07-31 | Marked: 2026-07-31

SECTION A (MCQS)	SECTION B (SHORT)	SECTION C (LONG)	TOTAL	GRADE
12/12	33/33	19.5/20	64.5/65 (99.2%)	A+

DEEP-CHECK VERIFIED (independently re-checked twice). Two separate deep-check passes, each a full page-by-page re-derivation against the exam paper and answer key, both confirm a single deduction on the entire 65-mark paper: half a mark on Q.3(iv)OR(b) for omitting the causal “why” behind enzyme optimum pH (see deduction log below). Both passes reconcile to the identical 64.5/65 total, with no further discrepancies found. This is Biology’s best-ever score on a fresh, previously-unseen full-syllabus Ch1-7 paper — and its 2nd-highest score of any kind, only half a mark short of a fifth perfect 65/65. Section B is a flawless 33/33, Biology’s first perfect Section B on a genuinely new (non-resit) paper.

SECTION A: Multiple Choice Questions — Score: 12/12

#	Question summary	Correct	Seemab	Result	Marks
1	Most important step of the biological method	C: Experimentation	C	Correct	1
2	Characteristic an unfalsifiable statement fails	D: Testable & falsifiable	D	Correct	1
3	Prions are composed of	B: Proteins only	B	Correct	1
4	Unicellular prokaryotes placed in kingdom	C: Monera	C	Correct	1
5	NOT present in a prokaryotic cell	D: Membrane-bound nucleus	D	Correct	1
6	Bivalents line up at the equatorial plate in	B: Metaphase I	B	Correct	1
7	Tumour of normal cells that stays in place	C: Benign	C	Correct	1
8	Two+ tissue types working together form an	B: Organ	B	Correct	1
9	Plant tissue that covers and protects	C: Dermal tissue	C	Correct	1
10	Disaccharide present in milk	C: Lactose	C	Correct	1
11	Copying a DNA base sequence into mRNA	B: Transcription	B	Correct	1
12	Detachable inorganic ion needed by an enzyme	A: Activator	A	Correct	1

Section A: 12/12. Seemab recorded her MCQ answers as a plain numbered list ("MCQs") on the final page of her scan rather than a bubble panel; all twelve letters are clearly legible and every one matches the answer key exactly.

SECTION B: Short Questions — Score: 33/33

Q.2(i), MAIN — Four characteristics a hypothesis should have 3/3

"It should be a general statement. It should be a tentative idea. It should agree with available observations. It should be testable and potentially falsifiable."

All four required characteristics present, matching the answer key exactly. Full marks.

Q.2(ii), MAIN — Two living + two non-living virus characteristics, and why not in any domain/kingdom 3/3

"Living: they occur in different varieties; they have their own genetic material in the form of either RNA or DNA. Non-living: they do not respire; they can be crystallized or stored in a bottle; they lack cellular structure and an enzyme system. Viruses are at the borderline of living and non-living characteristics, so they aren't included in any domain or kingdom under modern classification."

Both characteristic pairs correct (a bonus third non-living trait given), with the correct explanation for why viruses sit outside all classification. Full marks.

Q.2(iii), OR — Define saprotroph, autotroph, heterotroph + a Eukarya kingdom for each 3/3

"Autotrophs: organisms capable of producing their own food (photosynthetic mode of nutrition) — Kingdom Plantae, producers. Heterotrophs: organisms which eat other things as food (ingestive mode of nutrition) — Kingdom Animalia, consumers. Saprotrophs: organisms which depend on dead, decaying matter (absorptive mode of nutrition) — Example: Fungi, Bacteria; decomposers."

All three definitions are correct. For the required "one Eukarya kingdom" per category, Seemab named Fungi correctly for saprotrophs (listed first, and on its own already sufficient for full marks), Animalia for heterotrophs and Plantae for autotrophs. She additionally wrote "Bacteria" alongside Fungi as a second example for saprotrophs — Bacteria decomposes dead matter too, but it is neither a kingdom nor a member of domain Eukarya (it is prokaryotic, its own separate domain). Since only one correct kingdom was required and Fungi was given first, the mark stands at 3/3, but this mix-up is flagged under Areas to Revise below.

Q.2(iv), MAIN — Three muscle cell types, location of each, contractile protein 3/3

"Smooth muscle: in the throat and in the wall of the gut, producing the peristaltic contraction that moves food along. Striated muscle: in the muscles of arms and legs, which move body parts. Cardiac muscle: in the heart. Muscle cells have an elongated shape and are filled with actin and its associated contractile proteins, which is what allows them to contract and relax."

All three types, their locations, and the contractile protein are all correct, matching the answer key almost word for word. Full marks.

Q.2(v), OR — Metaphase I and anaphase I; chromosomes per pole when 2n=12**3/3**

"Metaphase I: the bivalents move to the equatorial plate of the spindle; one member of each homologous pair is attached to a spindle fibre from one pole, the other from the opposite pole. Anaphase I: the two homologous chromosomes of each pair separate and move to opposite poles; each chromosome still has 2 chromatids, each pole receiving a haploid set. If the parent cell has 2n=12, the haploid number is 6, so each pole receives 6 chromosomes at the end of anaphase I."

Both phases correctly described and the chromosome count correctly derived from 2n=12. Full marks. Note: she writes "equilatoral plate" here rather than "equatorial plate" — this reads as a terminology/spelling slip rather than a conceptual mix-up, since she uses the correct term, "the equator of the spindle," for Metaphase II a few pages later in Q.3(ii). Flagged under Areas to Revise below.

Q.2(vi), MAIN — Identify stages A-E on the mitosis figure (Fig 4.3)**3/3**

"E: Interphase. D: Prophase. A: Metaphase. B: Anaphase. C: Telophase."

All five labels correct, matching the answer key exactly. Full marks.

Q.2(vii), OR — Four types of animal tissue + one function of each**3/3**

"Muscle tissue: contracts and relaxes to produce movement of the body and of materials inside it. Epithelial tissue: covers and lines body surfaces and organ surfaces, giving protection and forming a barrier. Nerve tissue: conducts nerve impulses, carrying out coordination and control. Connective tissue: joins, supports and binds other tissues and organs together."

All four tissue types named with the correct function for each. Full marks.

Q.2(viii), MAIN — Three plant tissue types + function of each**3/3**

"Dermal tissue: covers and protects the plant, forming its outer covering. Ground tissue: site of photosynthesis, also stores food and provides support. Vascular tissue: transports material through the plant; xylem carries water and dissolved minerals, phloem carries prepared food."

All three tissue types and functions correct. Full marks.

Q.2(ix), MAIN — Monosaccharide units of sucrose/maltose/lactose + what else is released**3/3**

"(a) Sucrose = glucose + fructose: $C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$. (b) Maltose = glucose + glucose. (c) Lactose = galactose + glucose."

All three correct; the water-release note is shown once on the sucrose equation, which the mark scheme accepts as covering all three. Full marks.

Q.2(x), OR — Describe translation: location, role of tRNA, product**3/3**

"Translation takes place in the cytoplasm. The mRNA synthesized in the nucleus moves out into the cytoplasm, where it binds to a ribosome. Role of tRNA: it transfers amino acids to the ribosome according to the sequence of codes on the mRNA; different types of tRNA carry different amino acids, so the right amino acid arrives for each code. The ribosome reads the code carried by the mRNA and links the amino acids together, producing a polypeptide chain with a specific sequence of amino acids, that is a protein."

Location, tRNA role, and product are all correctly described. Full marks.

Q.2(xi), MAIN — Three types of co-factor + one example of each**3/3**

"A co-factor is the non-protein component that many enzymes need to function properly. Activator: a detachable inorganic ion, e.g. Zn^{++} , Fe^{++} , chloride ions. Prosthetic group: an organic molecule tightly or permanently attached to the enzyme, e.g. FAD, haem group. Coenzyme: a detachable organic molecule, e.g. NAD, vitamin A and C."

All three co-factor types correctly defined with a valid example each. Full marks.

Section B: 33/33 — a perfect score. This is Biology's first flawless Section B on a genuinely new (non-resit) full-syllabus paper.

SECTION C: Long Questions — Score: 19.5/20**Q.3(i), OR — (a) Four hypothesis characteristics (b) deduction + worked example (c) why experimentation is most important****5/5**

"(a) A hypothesis should be a general statement, a tentative idea; it should agree with available observations; it should be testable and potentially falsifiable. (b) A deduction is a logical consequence of a hypothesis: the hypothesis is taken as true and the observer works out what must follow from it using 'if...then' logic. Worked example — hypothesis: 'malaria is caused by Plasmodium.' Deduction: 'if malaria is caused by Plasmodium, then all malarial patients should have Plasmodium in their blood.' (c) Experimentation is the most important step because it is the only step that actually settles whether the hypothesis is true or false; the deductions drawn from it are subjected to rigorous testing, and it is through experimentation that a biologist learns which hypothesis is correct — without it a hypothesis stays no more than a plausible guess."

All three parts fully correct and matching the answer key's reasoning almost word for word, with a valid worked if-then example built around a real pathogen. Full marks.

Q.3(ii), MAIN — Four stages of meiosis II, daughter cells, interphase II**5/5**

"Prophase II: chromatin coils up, chromosomes become visible again, spindle apparatus organizes, nuclear envelope disintegrates. Metaphase II: chromosomes line up separately (not in homologous pairs) at the equator, each attached by a spindle fibre from both poles. Anaphase II: the centromere divides and sister chromatids are pulled apart to opposite poles. Telophase II: chromatids reach the poles, uncoil into chromatin fibres, a nuclear envelope forms around each set, cytokinesis II divides the cytoplasm. At the end of meiosis II there are 4 daughter cells, haploid (n); because of crossing over in prophase I, they aren't genetically identical to each other. During interphase II, no DNA replication takes place — the genome was already replicated in interphase I."

All four stages, the daughter-cell count/ploidy/genetic-difference reasoning, and the interphase II point are all correct and complete. Full marks.

Q.3(iii), MAIN — DNA to protein flow: the two stages, where each occurs, the RNA types

5/5

"DNA (in its genes) contains the information needed for protein synthesis. The flow of information from DNA to RNA and then RNA to protein is known as the central dogma of molecular biology; protein synthesis takes place in two sequential steps, transcription and translation, together called gene expression. Transcription: both DNA strands unwind, and the nucleotide sequence of only the coding strand is copied into an mRNA molecule; the DNA strands then re-coil. Translation: the mRNA moves into the cytoplasm and binds a ribosome, which reads the code and links amino acids together in sequence, building a polypeptide chain. Transcription occurs in the nucleus; translation occurs in the cytoplasm, at the ribosome. mRNA carries the message of DNA from the nucleus to the cytoplasm for protein synthesis; tRNA transfers amino acids to the ribosome as per the sequence of codes in the mRNA."

Every required part — information flow, both stages, both locations, and both RNA types — is present and correct. Full marks.

Q.3(iv), OR — (a) enzyme specificity (b) optimum pH of pepsin/trypsin (c) effect of high/low temperature

4.5/5

"(a) An enzyme is specific because it catalyzes only one type of reaction and acts on only one particular substrate; the reason is the active site, a cavity of a specific shape/charge that only a matching substrate can bind. Example: amylase acts on starch only, not on proteins or fats. (b) Pepsin has an optimum pH of 2.00, works best in the stomach's acidic medium; trypsin's optimum pH is 8.00, works best in the small intestine's alkaline medium. (c) Far above the optimum, the enzyme's atoms vibrate so vigorously that the bonds holding its globular structure break; the globular structure and active site are permanently damaged — denaturation. Far below the optimum, the enzyme is merely inactive because there isn't enough activation energy for the reaction to proceed; its structure is undamaged, so it regains activity as soon as the normal temperature is restored."

Part (a) specificity and part (c) temperature effects are both textbook-precise and complete. Full marks on those two parts.

Part (b): Seemab gets the descriptive core of the answer entirely right — both correct optimum pH values (pepsin 2.00, trypsin 8.00), both correct working locations (stomach, small intestine), and the acidic-versus-alkaline contrast between them are all present. What's missing is one layer deeper: the causal mechanism the question is really asking for, i.e. that an enzyme's active site only holds its correct shape at its own optimum pH, which is why pepsin and trypsin need such different values. —0.5 of the 2 marks for this part, kept consistent with how this identical sub-topic (pepsin/trypsin optimum pH) was marked on exam-12's Q2(ix)/Q3(iv) — the 2nd occurrence of this specific missing-depth pattern.

Deduction Log (reconciles to headline total)

Question	Marks available	Marks lost	Reason	Marks awarded
Section A, Q.1-12	12	0	All twelve MCQs correct (recorded as a numbered list on the final scan page, all legible)	12
Q.2(i)-Q.2(xi)	33	0	All eleven Section B questions fully correct, matching the answer key precisely	33
Q.3(i)OR, Q.3(ii), Q.3(iii)	15	0	All three fully correct with complete working and reasoning	15
Q.3(iv)OR	5	0.5	Specificity and temperature parts fully correct; part (b) gives correct pH values, locations and the acid/alkaline contrast, but doesn't go one layer deeper into the active-site-shape mechanism behind why the two values differ	4.5
Total	65	0.5	Reconciles exactly to headline	64.5

Strengths

- Perfect Section A (12/12): every MCQ correct, again recorded as a numbered list on the last scan page rather than a bubble panel, fully legible and verified.
- Perfect Section B (33/33): Biology's first flawless Section B on a fresh, non-resit paper — all 11 short questions scored full marks with textbook-precise wording.
- Section C nearly perfect (19.5/20, 97.5%): the biological-method worked example (malaria/Plasmodium), the four stages of meiosis II, and the full DNA-to-protein central dogma explanation were all flawless.
- 2n=12 → haploid=6 chromosome derivation correctly worked (Q2(v)OR).
- All five mitosis-figure stage labels (A–E) correctly identified (Q2(vi)).
- Correct sucrose/maltose/lactose monosaccharide breakdown with a balanced condensation equation (Q2(ix)).
- Enzyme specificity explained with the full active-site mechanism and a concrete worked example (Q3(iv)OR(a)).
- Correct use of "equator of the spindle" for Metaphase II (Q3(ii)) — shows the underlying concept is solid; see the terminology note opposite for the one place the wording slipped.
- 99.2% overall — Biology's best-ever score on a fresh, previously-unseen full-syllabus Ch1-7 paper, and its 2nd-highest score of any kind, half a mark short of a perfect 65/65.

Areas to Revise

- Enzyme optimum-pH "why" reasoning (Q3(iv)OR(b)): correct pH values (2.00/8.00), locations, and acid/alkaline contrast all given — only the deeper mechanism is missing, that each enzyme's active site is shaped to work only at its own environment's pH.
- This is now the 2nd occurrence of this exact gap on this exact sub-topic (pepsin/trypsin optimum pH), also seen on exam-12's Q2(ix)/Q3(iv) — worth a dedicated flash-card drill on "why enzymes have an optimum pH."
- Terminology slip: "equatorial plate" instead of "equatorial plate" (Q.2(v)OR). Marks unaffected (3/3) and this is not a conceptual error — she correctly writes "the equator of the spindle" for Metaphase II a few pages later (Q.3(ii)) — but the spelling is worth tidying up before it reads as a mistake in a board exam.
- Domain/kingdom mix-up: Q.2(iii)OR named Fungi correctly as a Eukarya kingdom for saprotrophs, then added Bacteria alongside it. Bacteria is neither a kingdom nor part of domain Eukarya — it is prokaryotic and forms its own separate domain. Marks unaffected (3/3, since only one kingdom was required and Fungi was listed first), but worth a quick review of the domain-vs-kingdom hierarchy so Bacteria isn't offered as a Eukarya kingdom again.
- No other mark-losing errors found anywhere on the paper — this is otherwise Seemab's cleanest Biology paper to date, and the first Biology exam in five attempts (since exam-09) with zero instances of the "stopping before completing every required part" pattern.

Comparison vs Previous Biology Attempts

Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
3 Apr 2026	exam-01-ch1-6	11/12	28/33	13/20	52/65	80.0%	A
30 Apr 2026	exam-02-ch7 (1st)	12/12	31/33	17.5/20	60.5/65	93.1%	A
2 May 2026	exam-02-ch7 (re-sit)	12/12	33/33	20/20	65/65	100%	A+
8 Jun 2026	exam-03-ch4	12/12	31/33	19.5/20	62.5/65	96.2%	A+
23 Jun 2026	exam-07-ch1-7	12/12	27/33	18.5/20	57.5/65	88.5%	A
29 Jun 2026	exam-08-ch1-7	12/12	25/33	17.5/20	54.5/65	83.8%	A

6 Jul 2026	exam-09-ch1-7	12/12	32/33	17.25/20	61.25/65	94.2%	A+
16 Jul 2026	exam-11-ch1-7	10/12	26.9/33	19.5/20	56.4/65	86.8%	A
22 Jul 2026	exam-12-ch1-7	12/12	29/33	19.3/20	60.3/65	92.8%	A+
31 Jul 2026	exam-13-ch1-7 (this exam)	12/12	33/33	19.5/20	64.5/65	99.2%	A+

A fresh, previously-unseen full-syllabus Ch1-7 paper (exam-13, Version 1). At 99.2% this is Biology's best-ever score on a first-sitting paper, and its 2nd-highest score of any kind (behind only the 100% re-sit of the already-seen exam-02 paper). Section B is a flawless 33/33 for the first time on a genuinely new paper. Section C narrowly misses full marks (19.5/20) on a single half-mark deduction: the enzyme optimum-pH question gave correct values, locations and contrast but not the deeper active-site-shape reasoning behind them — the 2nd time this exact gap has appeared (also exam-12). Biology's five full-syllabus Ch1-7 papers now rank: exam-13 99.2%, exam-09 94.2%, exam-12 92.8%, exam-07 88.5%, exam-11 86.8%, exam-08 83.8%.